

DIRECTED ENERGY WEAPONS: High-Power Microwaves



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In the previous parts of this article we talked about particle beam weapons and their technology, in addition to touching base on high-power microwaves (HPMs). In this part, we learn more about HPMs.

e-Bomb

An electromagnetic bomb, or e-bomb, makes use of a short, intense electromagnetic energy pulse, adversely affecting electronics circuitry with practically no effect to humans or buildings. The damage may range from temporarily disabling electronics systems functioning at low levels of electromagnetic pulse (EMP) energy to corrupting computer data at mid-range levels and further to complete destruction of electronic circuitry at high levels.

EMP effect first observed during the early testing of high-altitude airburst nuclear weapons is characterised by production of a very short, typically hundreds of nanoseconds wide, intense electromagnetic shockwave travelling away from the source. This short and intense magnetic field is capable of producing short-lived transient voltages of kilo-volt level on exposed electrical conductors including wires or conducting tracks on printed circuit boards. This makes it militarily significant as it can inflict irreversible damage to a wide range of electronics and computer systems in command, control, communications hardware and radar systems.

Radar and electronic warfare equipment, satellites, microwaves, ultra-high-

frequency, very-high-frequency, high-frequency and low-band communications equipment and television equipment are all potentially vulnerable to EMP effect. The inflicted damage is similar to that experienced through exposure to close-proximity lightning strikes, and may require complete replacement of the equipment, or at least substantial portions thereof.

Commercial computer equipment is particularly vulnerable to EMP effects due to large-scale use of high-density metal oxide semiconductor devices that are highly sensitive to high-voltage transient exposure. Shielding provided by equipment chassis in this case gives only limited protection as the high-voltage transients find their way into the equipment through input/output cables. Computers used in data processing systems and communication systems and those embedded in military equipment are all potentially vulnerable to EMP effect.

Key technologies involved in generation of high-power EMP, which mainly include explosively-pumped flux compression generator (FCG), explosive or propellant-driven magnetic-hydrodynamic (MHD) generator and virtual cathode oscillator (vircator), have matured to an extent that realisability of battlefield deployable e-bombs has become feasible. This has opened up many a new application in both tactical and strategic information warfare.

Development of conventional e-bomb devices has also allowed their use in non-nuclear confrontations. In the following paragraphs are discussed operational principle of e-bombs, lethality of e-bombs and their delivery mechanisms. A brief reference is made to some of the experiments carried out with e-bombs.

Operational principle

The explosively-pumped FCG capable of producing tens of mega-joules of electri-

*Fig. 6: FCG based
low-frequency e-bomb*

